

Simple Rocket Motor Grain Burnback Regression with Fast Marching vs. Level Set

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Abstract

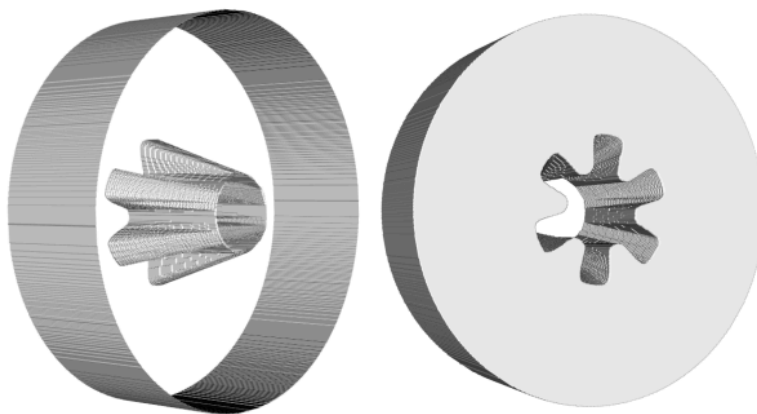
This paper compares the Fast Marching Method (FMM) and the Level Set Method (LSM) for modelling the burnback regression of solid rocket motor (SRM) propellant. An SRM keeps propellant in a solid state, completing the burn from inside out. To simulate the regression, FMM solves the Eikonal equation using a single-pass shortest path algorithm. The method explicitly holds arrival times for each point in a 3D grid, producing fast results for uniform burn conditions. Without the ability to resolve merging points or dynamic burn rates, the algorithm breaks down under erosivity and topology changes. LSM overcomes these complications by evolving a signed distance field through time. Implicit modeling with the zero level set displays the surface at any point in time. Burn rate recalculations occur at every step using the Hamilton-Jacobi equation, Godunov's upwind scheme, and TVD Runge-Kutta stabilization. Compared with the Fast Marching Method (FMM), the Level Set Method (LSM) uses a more accurate and adaptable voxel (3-Dimensional pixel) based algorithm for modelling 3D grain surface regression, yielding improved performance and increased safety for solid propulsion systems. As the aerospace industry adopts digital twin frameworks, LSM's coupling capability proves foundational for next-generation propulsion simulation.

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Humans last walked on the moon in December 1972. Six decades later, NASA expects humans to walk on Mars. Increased aspiration to settle the world beyond current reach has heightened the importance of reliable and powerful propulsion systems. Solid rocket motors (SRMs) use propellant kept in a solid state, as seen in Figure 1 (Liu et al., 2023, p. 3), enabling easy storage and use. Difficulties often arise when determining the behavior of a specific propellant shape (grain geometry) during regression, making precise modelling methods key to a successful launch. Compared with the Fast Marching Method (FMM), the Level Set Method (LSM) uses a more accurate and adaptable voxel (3-Dimensional pixel) based algorithm for modelling 3D grain surface regression, yielding improved performance and increased safety for solid propulsion systems. FMM follows a static framework like Dijkstra's by computing a single-pass arrival time, making the simulation faster, but less precise for complex grain geometries. LSM models regression by computing distance continuously through time, capturing the geometric and physical changes FMM struggles with. The foundations of FMM show why these limitations are mathematical rather than flaws in code.

Figure 1

Solid rocket motor grain reconstruction using a CT image



Fast Marching Method Technical Background

Rooted in the mathematics of wave propagation and shortest path computation, FMM solves problems with moving boundaries, such as melting snow, seismic wave travel, robotic vision, and medical imaging. The speed function F of a regression stays positive, meaning that the boundary of an object or wave continues moving outwards at a fixed speed. In ideal cases, SRM propellant burning follows these exact characteristics. As a result, FMM treats grain burnback as a propagating surface moving through a grid, computing arrival times at each point. The grid setup defines how those arrival times get computed.

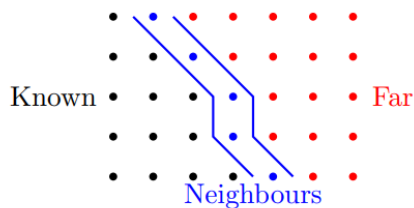
FMM begins by discretizing the grain geometry into a 3D grid with voxel height h . A status matrix tracks each point with one of three values. A zero marks trial points, one to accepted points, and negative one to far points. Another matrix with the same size simultaneously stores the arrival time for each point. At simulation start, surface points default to time zero, and their neighbors enter the trial group. For each trial point, a quadratic formula (1) computes an arrival time using the shortest known neighbor times on all three axes, approximating a discretized version of the Eikonal equation (2). Figure 2 (Alblas, 2018, p. 5) represents how the trial wave propagates through the grid as the simulation advances.

$$(T - T_{xmin})^2 + (T - T_{ymin})^2 + (T - T_{zmin})^2 = h^2/F^2 \quad (1)$$

$$|\Delta T| = 1/F \quad (2)$$

Figure 2

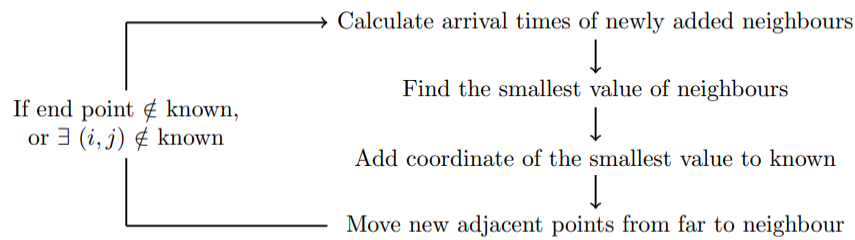
The Known, Neighbor, and Far Points on a 2D Grid



As the wave propagates, trial point times enter a min-heap tree, which holds the smallest element at the root and keeps every child larger than its parent. At each step, the algorithm extracts the smallest node, inserting it into the time matrix. The accepted node's far neighbors then enter trial and get inserted into the bottom of the tree. The algorithm percolates the inserted node up when its parent holds a larger value. Figure 3 (Alblas, 2018, p. 5) outlines the algorithm's flow chart. The algorithm completes once both matrices hold a value for every point, allowing burn surface reconstruction at any time stamp.

Figure 3

Flow Chart of FMM



Bhattacharya and Mundhra (2025), two mechanical engineers researching propulsion systems and fluid mechanics in India, validated FMM's algorithm by modeling regression with 2D splines. In an ideal case, a uniform object regresses identically across its entire length, allowing the FMM cross-sectional analysis to work. Shortest-path algorithms model 2D splines effectively, making complex-geometry simulation feasible. Bhattacharya and Mundhra highlight that this dimensional reduction matters because the more complex a fractal grain becomes, the more efficiently a motor will burn (p. 9). Table 1 (Bhattacharya & Mundhra, 2025, p. 4) shows notable impulse and burn-time errors when modeling regression with FMM, revealing two key weaknesses under non-idealized SRM conditions: topology change and erosive burning. FMM handles two waves reaching the same point by taking the one that reached it first, despite both

waves acting on that point simultaneously. The error compounds with every incorrect topology change as grain complexity increases. Erosivity produces burn-rate gradients due to non-uniform gas distribution along the surface, which violates FMM's static framework. LSM removes both limitations by computing a continuously evolving boundary through time.

Table 1

Comparison Between Actual Motor Test and Simulated Motor Test using OpenMotor

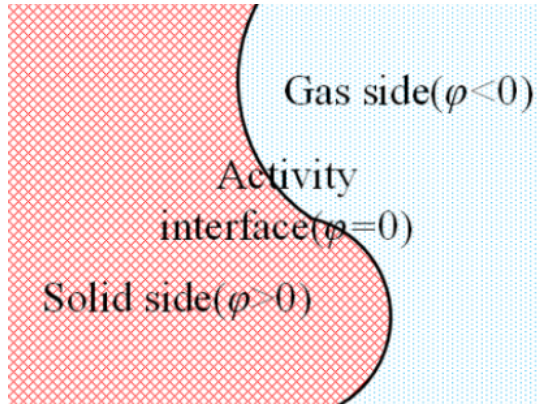
Parameter	Actual motor	Simulated motor	Percentage difference
Burn time/action time, s	34.4	35.5	+3.2%
Propellant weight, kg	333.6	331.2	−0.73%
Action time, s	37.3	39.09	+4.8%
Burn time averaged Kn	233.56	234.38	0.35%
Action time average chamber pressure, MPa	3.461	3.447	−0.4%
Maximum chamber pressure, MPa	3.923	3.978	+1.40%
Total impulse, kN-s	951.3	916.2	−3.69%
Delivered specific impulse, N-s/kg	2820.65	2795.5	−0.83%

Support

The level set method approaches grain burnback regression as a continuously evolving boundary problem by assigning voxels in the 3D grid a signed distance from the surface. Unburned voxels carry positive values, burned voxels carry negative values, and surface voxels carry zero. The center of the voxel stores the distance. Across the entire grid, these values form the signed distance field (SDF) ϕ (phi) as seen in Figure 4 (Liu et al., 2023, p. 5). LSM implicitly models the moving surface by extracting where ϕ is zero (zero level set) rather than storing the surface at every time. Maintaining a coherent grid across the entire simulation resolves complex topology changes. LSM's algorithm recalculates F at every step, yielding accurate results under erosive burning. LSM captures real-life properties during simulation produced by the specific implementation design.

Figure 4

Signed Distance Field



Level Set Method Technical Detail

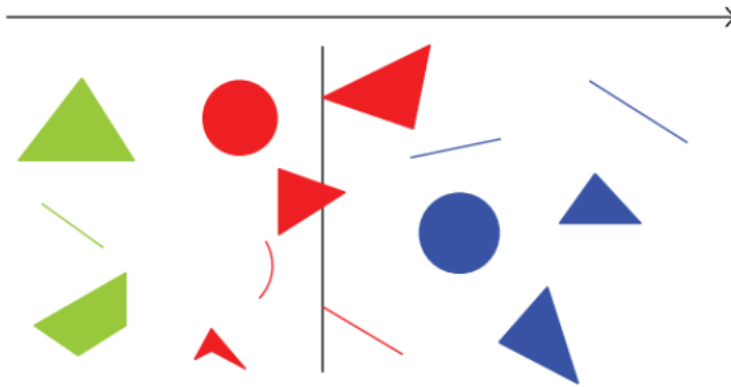
Before beginning, the simulation initializes a 3D voxel grid alongside two matrices of equal size. Each voxel receives a coordinate (i,j,k) holding a ϕ value in one matrix and an F value in the other. Each 3D matrix flattens into a single-dimensional array with specific indexing math to find each coordinate value. To generate the initial SDF, each voxel needs a minimum distance field (MDF) and a sign.

Wei et al. (2018), engineers at Northwestern Polytechnical University in the Science and Technology on Combustion Internal Flow and Thermal-Structure division, implement an efficient scanning-based method for SDF initialization called DiFi. Wei et al. note that brute forcing calculations for each voxel requires hours of operation, whereas a method for using smaller local subsets proves more efficient (p. 3). Difi sweeps the grain in one direction with a line, splitting elements into three groups. Figure 5 (Wei et al., 2018, p. 4) shows the already swept, active, and approaching groups in green, red, and blue, respectively. To calculate a voxel's minimum distance, only the nearest neighbors need to contribute. The propagation runs in both directions to confirm the shortest distance. Wei et al. prove that restricting computation to the red group significantly reduces processing time (p. 3). A ray cast from each voxel checks for

boundary intersection, determining the point's position relative to the grain. After initializing the SDF matrix, the algorithm must calculate the speed field.

Figure 5

DiFi Demonstration



Vielle's burn rate law (3) helps calculate F for each voxel using specific constants a and n defined by the propellant chemistry. P defines the value of local chamber pressure.

$$F = aP^n \quad (3)$$

At the beginning, the F matrix gets assigned base values because of uniform P . The F matrix only becomes spatially variable once the simulation starts running and the flow of gas produces different velocities. Liu et al. (2023), engineers at the School of Aircraft Engineering and Nanchang Hangkong University, study surface regression under erosive burning. Liu et al. argue that properly modelling the regression of a complex SRM requires non-uniform gas distribution calculations (p. 1). The Lenoir and Robillard model computes the corrected F to account for erosive burning using equation 4.

$$r = ap^n + \alpha (G^{0.8} / D^{0.2}) e^{(-\beta r_{0pc}/G)} \quad (4)$$

G and D govern erosive burn rate since they change at every time step. These variables define the amount of gas moving across a particular point (gas mass flux) and the diameter of the port

(area through which the gas moves). The rest of the coefficients derived from thermal properties stay constant throughout the simulation (Liu et al., 2023, p. 4). The corrected F gets re-inserted into the matrix after every time step when the simulation begins.

After the simulation begins, surface movement determines the updates on each voxel (Liu et al., 2023, p. 5). The continuous Hamilton-Jacobi level set equation (5) governs the evolution of the burn surface.

$$\partial \phi / \partial t + F|\Delta \phi| = 0 \quad (5)$$

The partial derivative of ϕ describes how ϕ changes at the given time on a specific voxel. The gradient magnitude $|\Delta \phi|$ encodes the direction of surface movement. Liu et al. (2023) discuss a common method for approximating $|\Delta \phi|$ in the discretized case using Godunov's scheme on pages 5-6. Godunov's scheme computes the forward and backward finite differences of each axis using equations 6 and 7.

$$D^{-x}_i \phi = (\phi_i - \phi_{i-1}) / h \quad (6)$$

$$D^{+x}_i \phi = (\phi_{i+1} - \phi_i) / h \quad (7)$$

Each comparison produces a slope measuring how steeply ϕ changes between neighboring voxels. Equation 8 combines these slopes into the central difference Δ used by the Godunov scheme.

$$\begin{aligned} \nabla^+ = & (\max(D^{-x}, 0)^2 + \min(D^{+x}, 0)^2 + \\ & \max(D^{-y}, 0)^2 + \min(D^{+y}, 0)^2 + \\ & \max(D^{-z}, 0)^2 + \min(D^{+z}, 0)^2)^{1/2} \end{aligned} \quad (8)$$

The final update at a voxel (i, j, k) takes the discretized version of the Hamilton-Jacobi equation (9), where ϕ^n_i and ϕ^{n+1}_i represent the current and new value, respectively.

$$\phi^{n+1}_i = \phi^n_i - \Delta t F \nabla^+ \quad (9)$$

Thousands of small updates iterating through the simulation introduce a slight oscillating error, where ϕ overshoots its true value and overcorrects on the succeeding step. Liu et al. (2023) combat this drift through a TVD Runge-Kutta scheme (10) that runs equation 9 three times and combines intermediate ϕ values into a weighted average.

$$\phi^{n+1} = (1/3)\phi^{(n+1/2)} + (2/3)\phi^{(3)} \quad (10)$$

The stable ϕ^{n+1} then overrides the final F value in the matrix. Equation 10 only smooths the advancement of ϕ close to the zero level set because the sharpest changes happen there, leaving further ϕ values to accumulate estimation errors and slightly drift. Reinitialization fixes drift by re-running the SDF computation across the entire grid to confirm accurate distance from the zero level set at every voxel. Liu et al. (2023) handle reinitialization on pages 7-8 by checking whether a ϕ value differs from the true distance computed by the MDF, and updating the value accordingly. Even after obtaining an SDF, extracting the burn boundary into a visual model requires additional steps.

In practice, a singular ϕ never takes on the value zero. On a discrete grid, the surface always lies between adjacent positive and negative values, rather than at the voxel's midpoint. Marching cubes, a method Wei et al. (2018) incorporate, interpolate between voxels holding opposite signs. A triangular mesh then combines the surface points. As the simulation propagates, the mesh updates accordingly, displaying surface evolution under erosive burning and topology changes. Accurately extracting and updating the mesh at each step only matters if the simulation captures the physical properties that govern regression.

Analysis

The challenge in modelling an SRM regression comes down to the physical properties of how rockets burn. Grain geometry and burn rate create drastically different regression patterns

across the motor. Figure 6 (Liu et al., 2023, p. 12) exemplifies how increased burn rate, caused by changing chamber pressures, regresses one region at a different rate. Figure 7 (Liu et al., 2023, p. 10) shows how a simple grain will regress unevenly because changing mass flux creates erosive burning. FMM's biggest limitation was requiring that the burn rate stays unchanged. In practice, all SRMs experience erosive burning, which leads to topological changes even with simple grains. LSM overcomes this by continuously updating F . Burning surface area curves for star geometries produce measurably different results, as shown in Figure 8 (Liu et al., 2023, p. 12). LSM requires more computation than FMM, increasing run time significantly. Many researchers have addressed this cost through optimization.

Coupling with complementary methods such as Volume of Fluid (VOF), Boolean operations, and GPU parallelism magnifies LSM's strength. Since FMM computes once and never updates boundary conditions, coupling with a flow solver proves futile. LSM's continuously evolving field provides a boundary condition for coupled simulations. Zhang et al. (2021), engineers at the Air Force Engineering University in Xi'an, explore a coupled simulation with the VOF. Zhang et al. successfully overcame the shortcomings of calculating full transient internal flow (pressure, temperature, and velocity) since LSM provides a live boundary condition during every time step (p. 10). Wei et al. (2018) further explore optimization using GPU parallelism and Boolean operations to reduce run time by a factor of 15 and keep error under 0.25% (p. 10). These optimizations make LSM viable for industry use.

Figure 6

Regression With Differing Burning Rates

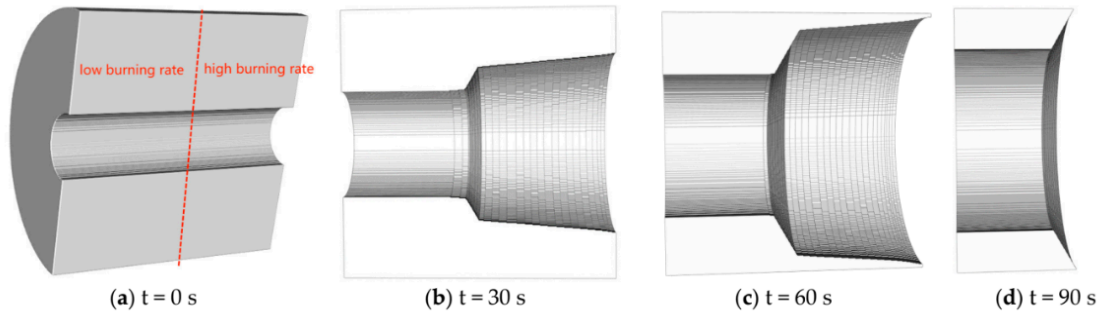


Figure 7

Erosive Burning on Uniform Geometry

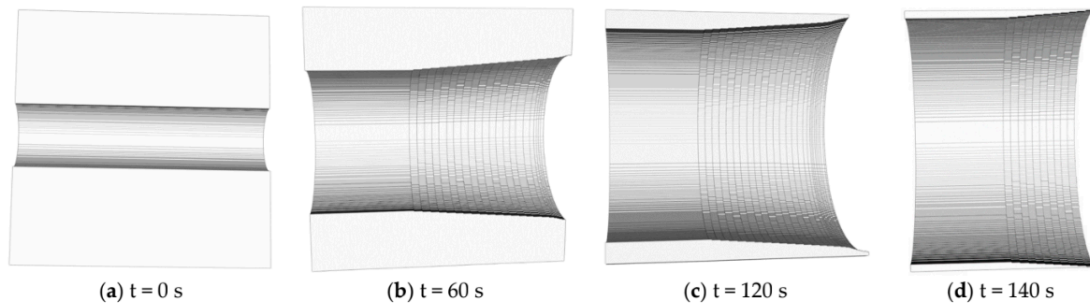
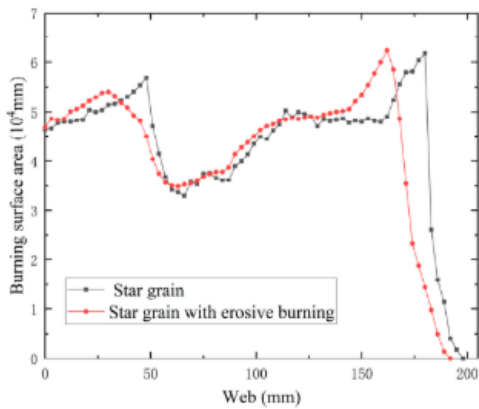


Figure 8

Comparison Between Star Grain Burning Surface Area With and Without Erosivity



Social Impact

The differences between how FMM and LSM handle grain burnback regression carry consequences well beyond accuracy. During the design of an SRM, engineers use simulations to

dictate how a specific motor may behave. Once the numbers meet expectations, that motor gets replicated during manufacturing. If a simulation underestimates burn time or misrepresents the thrust profile, the rocket could underperform during a critical mission. As a supplement to real-world physics, FMM proves unreliable due to its constant burn speed. In an industry where failures can destroy payloads worth hundreds of millions of dollars and endanger life, the choice of simulation becomes an ethical decision with direct implications on the real world.

Nancy Leveson (2020), a professor of Aeronautics and Astronautics at the Massachusetts Institute of Technology, addresses the relationship between simulations in technology and safety in her Communications of the ACM article. Leveson argues that reliability and safety represent two entirely different principles, and even a reliable system can produce harm (p. 2). Software carries implied harm when operators use the tool without properly understanding its intended applications. Leveson demonstrates the distinction with an example of firing a gun in an open field versus in a crowd of people (p. 2). The mechanism stays identical and produces the same shot every time, but the safety conditions differ vastly. Although FMM produces consistent results in lower-level fields where 3% error remains acceptable, use in industry-level missiles or rockets with lives at stake proves unsafe. Leveson further argues that accidents caused by software stem from incorrect models that can give false results or behave differently than expected (pp. 1-2). Simulations built on FMM give false results since they inaccurately display topological changes and erosive burning. Adopting LSM corrects the underlying model, addressing the root cause of inaccuracy in FMM.

Beyond correcting the model itself, LSM adapts to next-generation simulation tools. Karen Willcox (2023), an aerospace engineer and Director of the Oden Institute for Computational Engineering and Sciences at The University of Texas at Austin, explains digital

twins in her TED talk. A digital twin acts as a personalized, evolving model of a physical system (Willcox, 2023, 4:07). The strength of a digital twin comes from the ability to test anything before applying the results to a physical object. A simulation, on the other hand, serves as a broad estimate for any system. A digital twin assimilates data from the current state of an object to model it one-to-one in a virtual environment (Willcox, 2023, 1:33). As the aerospace industry moves towards digital twin modelling for launch vehicles, simulations must accept data at every time step to adjust predictions accordingly. LSM satisfies just that, with the ability to couple with other flow solvers and physics models. As grid resolutions increase and computers become faster, the gap between LSM simulation and physical motor behavior shrinks. That shrinking gap moves the aerospace industry closer to safer and more reliable production of propulsion systems.

Conclusion

Designing propulsion systems that perform safely under real flight conditions demands simulation tools capable of capturing the complexity of solid propellant burning. FMM's single-pass algorithm freezes each voxel permanently and assumes constant burn speed, producing measurable impulse and burn time errors. LSM eliminates both failure modes by evolving a signed distance field continuously, recalculating burn conditions at every step. Compared with the Fast Marching Method (FMM), the Level Set Method (LSM) uses a more accurate and adaptable voxel (3-Dimensional pixel) based algorithm for modelling 3D grain surface regression, yielding improved performance and increased safety for solid propulsion systems. As the aerospace industry expands and the number of space shots increases, accurate simulations become inseparable from human safety. The software professionals who select and validate simulations for propulsion devices carry direct responsibility for ensuring that safety-critical models reflect real-world conditions. Future research should focus on ways to

couple simulations and machine learning, as result-driven models can push the current limits of humanity. Artemis II proved that humans belong in space, but not without entrusting their lives to the simulations that got them there.

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